



ABC Analytics Pvt Ltd

Data Driven Decision with AI

AI Data Analysis Report

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Rows	84,550
Columns	61
Duplicate Rows	-
Version	1.0

Prepared by

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Table of Contents

1 Executive Summary

2 Dataset Overview

3 Analytical Highlights

4 Analysis Results

5 Overall Findings

6 Recommendations

7 Conclusion



Dataset Analysis Report

Executive Summary

This report provides a comprehensive analytical evaluation of a cybersecurity dataset consisting of 84,550 firewall traffic log records across 61 columns. The data was captured on June 6, 2025, during a high-density 20-minute window between 11:50:10 and 12:10:00. The traffic was monitored by a Palo Alto Networks PA-440 firewall device operating within the organization "Demo-IR-Org".

The primary objective of this analysis is to assess network traffic patterns, evaluate firewall rule enforcement, identify potential security risks, and detect operational inefficiencies. The findings reveal that while the vast majority of network connections were allowed, there are critical areas requiring immediate attention. These include a highly specific set of suspicious command-and-control (C2C) threat events, an over-reliance on broad firewall rules, and a severe data duplication issue where exactly 50% of the dataset consists of duplicate records.

Dataset Overview

The dataset provides a detailed log of network traffic passing through the organization's firewall. Below is a summary of the structural and quality metrics of the dataset:

1. Total Rows: 84,550
2. Total Columns: 61
3. Data Types:
4. Numerical: 19 columns (e.g., `bytes\_received`, `bytes\_sent`, `total\_bytes`, `pkts\_received`, `pkts\_sent`, `risk\_level`)
5. Categorical: 36 columns (e.g., `action`, `app\_category`, `applicaton`, `srcip`, `destip`, `rule\_name`)
6. Datetime: 5 columns (`Timestamp`, `start\_time`, `time\_created`, `time\_generated`, `timestamp`)
7. Boolean: 0 columns
8. Duplicate Rows: 42,225 rows (exactly 50.0% of the dataset)
9. Missing Values:
10. Threat-related columns (`\_threat\_category`, `\_threat\_family\_name`, `\_threat\_ioc`, `\_threat\_malware\_ip`, `\_threat\_source`, `\_threat\_type`, `\_threat\_victim\_ip`): 84,516 missing values (99.96% missingness).



11. Outbound Interface (`outbound\_if`): 4,990 missing values (5.9% missingness).

Statistical Findings

Key Traffic Metrics Summary

The table below outlines the key statistical metrics for network traffic volume, duration, and risk levels across the logged sessions:

Metric	Mean	Minimum	25%	50% (Median)	75%	Maximum	Standard Deviation
Bytes Received	30,977.90	0.0	0.0	74.0	256.0	104,054,808.0	1,088,820.12
Bytes Sent	5,712.05	60.0	86.0	106.0	1,230.0	54,619,888.0	357,447.56
Total Bytes	36,689.95	60.0	94.0	285.0	1,294.0	104,365,342.0	1,184,440.31
Packets Received	31.18	0.0	0.0	1.0	1.0	151,414.0	1,069.73
Packets Sent	17.42	1.0	1.0	1.0	3.0	208,181.0	1,048.14
Elapsed Time (sec)	14.32	0.0	0.0	0.0	0.0	8,455.0	135.98
Risk Level	2.54	1.0	1.0	3.0	3.0	5.0	1.10

Correlation Analysis

Strong linear relationships were identified among several key numerical variables:

- `bytes\_received` and `total\_bytes`: Very strong positive correlation of 0.954, indicating that received data constitutes the bulk of the total network traffic volume.
- `pkts\_received` and `total\_bytes`: Strong positive correlation of 0.871.
- `pkts\_sent` and `bytes\_sent`: Strong positive correlation of 0.865.
- `pkts\_received` and `bytes\_received`: Strong positive correlation of 0.833.
- `risk\_level` and `nat\_dest\_port`: Weak negative correlation of -0.256.

6. `risk\_level` and `destport`: Weak negative correlation of -0.200.

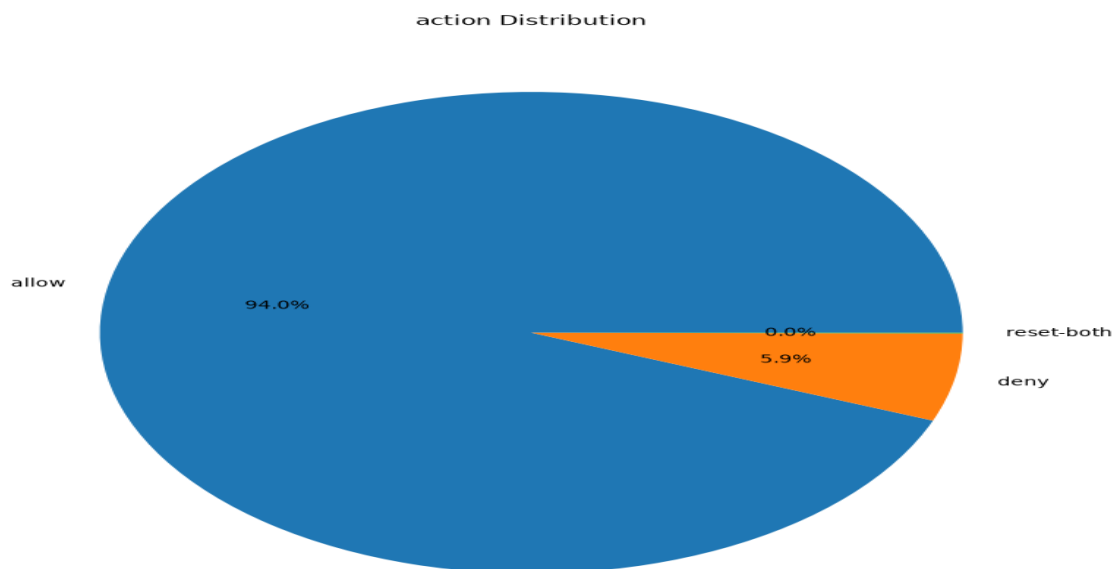
Data Anomalies and Outliers

The dataset contains a significant number of statistical outliers, indicating highly skewed distributions and occasional massive spikes in traffic volume and session duration:

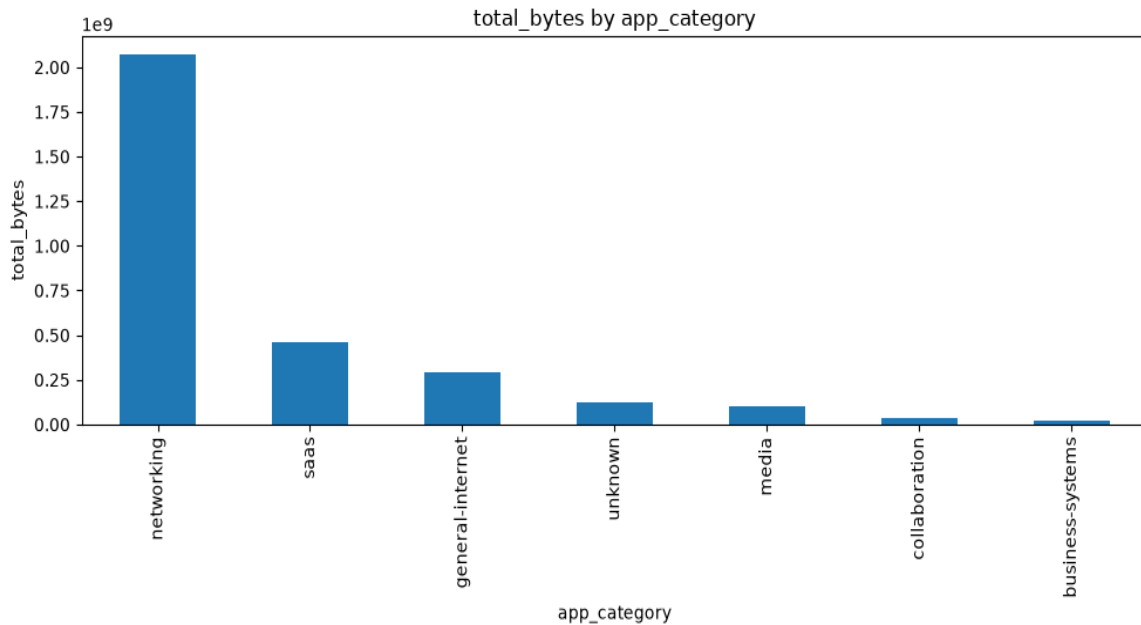
7. Packets Received (`pkts\_received`): 18,334 outliers
8. Elapsed Time (`elapsed\_time`): 17,446 outliers
9. Bytes Received (`bytes\_received`): 15,735 outliers
10. Packets Sent (`pkts\_sent`): 15,619 outliers
11. Total Bytes (`total\_bytes`): 12,291 outliers
12. Bytes Sent (`bytes\_sent`): 9,109 outliers
13. Destination Port (`destport`): 7,432 outliers
14. NAT Destination Port (`nat\_dest\_port`): 6,780 outliers
15. Session ID (`session\_id`): 10,050 outliers
16. Source Port (`srcport`): 1,098 outliers

Visual Analysis

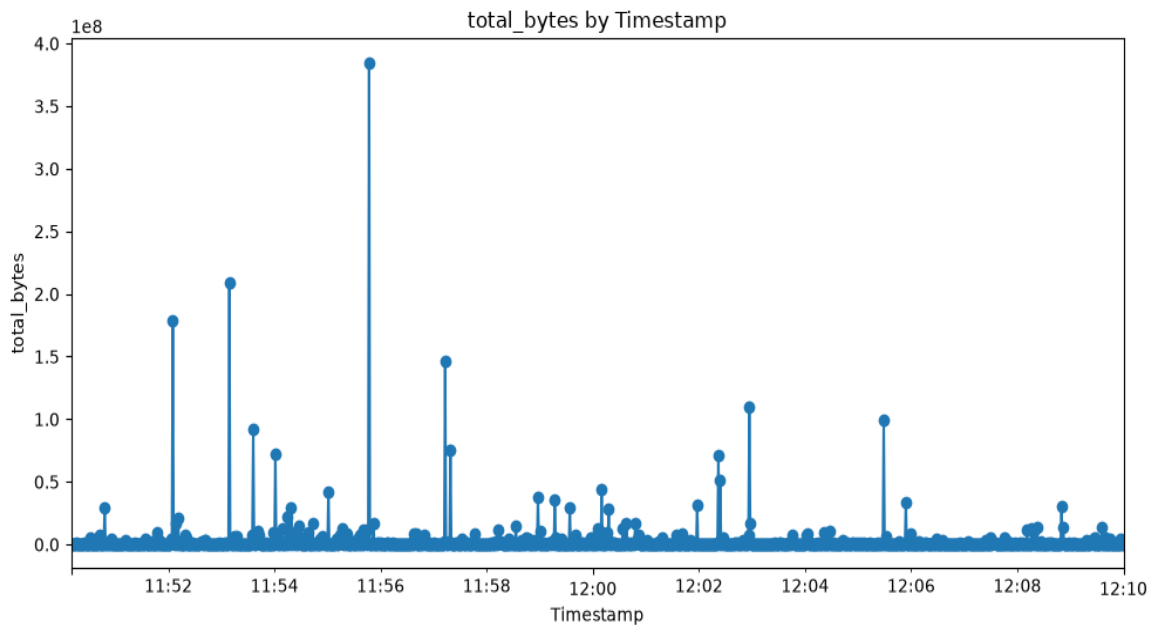
pie action



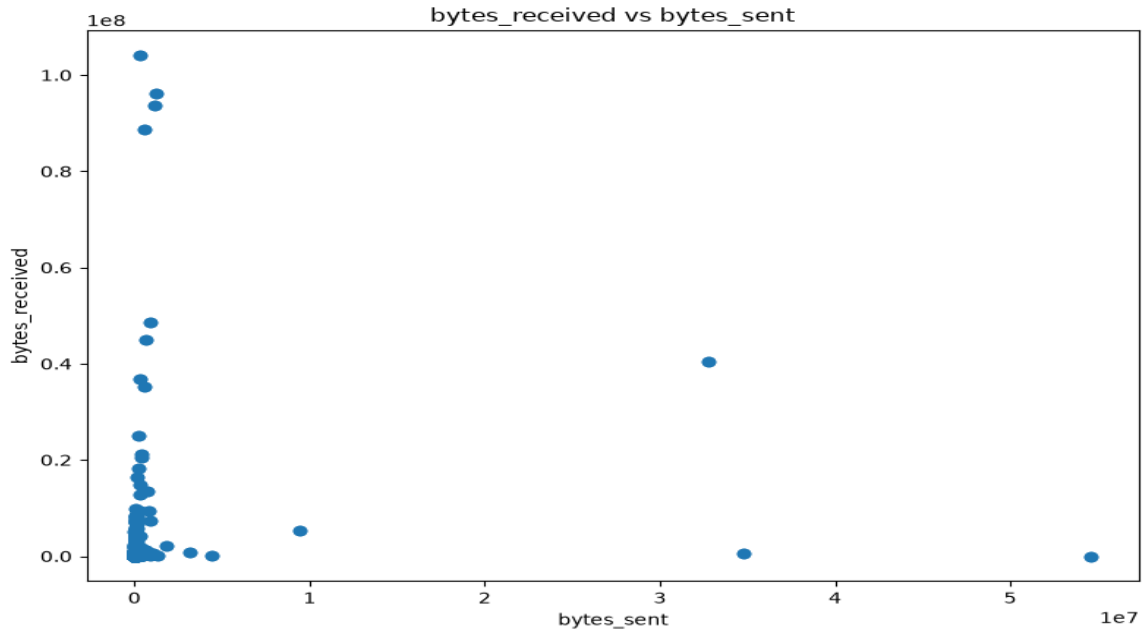
bar app category total bytes



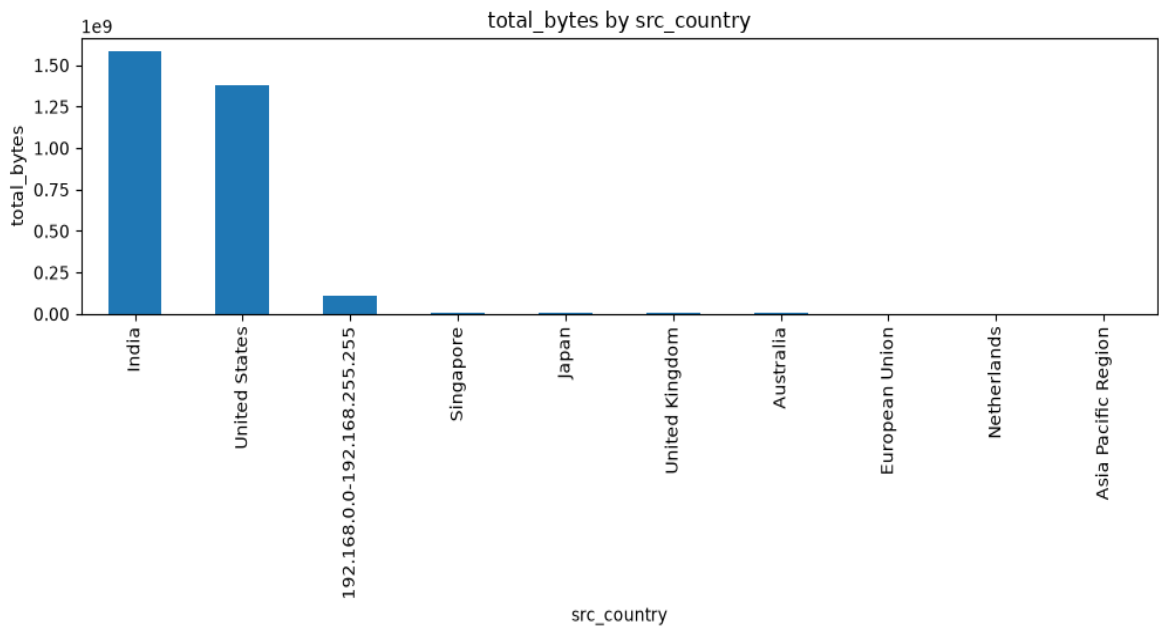
line Timestamp total bytes



scatter bytes sent bytes received



bar src country total bytes



Five visualizations were generated to analyze the network traffic and security logs:

1. Action Distribution Pie Chart (`outputs\charts\action\_pie.png`)

1. *Description*: This chart illustrates the percentage distribution of firewall actions across all logged connections.



2. ***Analysis*:** It visually highlights that the vast majority of traffic is allowed (94.05% / 79,518 connections), while a small portion is denied (5.90% / 4,990 connections), and an extremely minor fraction is reset (0.05% / 42 connections).

2. **Total Bytes by Application Category Bar Chart**
(`outputs\charts\app\_category\_total\_bytes\_bar.png`)

3. ***Description*:** This bar chart displays the total volume of data (in bytes) transferred across different application categories.

4. ***Analysis*:** It highlights the dominance of the `networking` category in terms of bandwidth consumption compared to other categories like `unknown`, `general-internet`, `saas`, and `collaboration`.

3. **Timeline of Total Bytes Line Chart** (`outputs\charts\Timestamp\_total\_bytes\_line.png`)

5. ***Description*:** This line chart plots the total bytes transferred over the exact timestamps of the 20-minute logging window.

6. ***Analysis*:** It shows the temporal distribution of network traffic, identifying specific high-density bursts and traffic spikes during the short monitoring period.

4. **Bytes Sent vs. Bytes Received Scatter Plot**
(`outputs\charts\bytes\_sent\_bytes\_received\_scatter.png`)

7. ***Description*:** This scatter plot maps individual sessions based on the volume of bytes sent (X-axis) versus bytes received (Y-axis).

8. ***Analysis*:** It illustrates the relationship between outbound and inbound data transfer sizes, visually separating typical low-volume sessions from extreme high-volume outliers.

5. **Total Bytes by Source Country Bar Chart** (`outputs\charts\src\_country\_total\_bytes\_bar.png`)

9. ***Description*:** This bar chart compares the total data volume (bytes) originating from different source countries.

10. ***Analysis*:** It confirms that the United States is the leading source of traffic volume, followed by India and Australia, matching the categorical distribution findings.

Business Insights

Important Patterns and Trends

1. **Application and Protocol Dominance:** The network is heavily dominated by infrastructure-level traffic. The primary application category is `networking` (61,661 occurrences), and the most frequent individual application is `dns-base` (38,076 occurrences). This aligns with the high usage of the UDP protocol (52,420 occurrences) compared to TCP (31,408 occurrences).

2. Destination Infrastructure: Network traffic is heavily directed toward public DNS resolvers, with Google's `8.8.8.8` receiving 17,873 connections and Cloudflare's `1.1.1.1` receiving 16,030 connections.

3. Geographic Concentration: The United States is the primary source of network traffic with 61,175 occurrences, followed by India (11,218) and Australia (8,051).

Risks

4. Policy Over-Permissiveness: The broad firewall rule `Allow-any` was triggered most frequently, accounting for 63,321 occurrences. This suggests that the firewall configuration may not be optimized for least-privilege access, exposing the network to unnecessary lateral movement risks.

5. Targeted Security Threat: Exactly 34 log entries (0.04% of the dataset) contain critical threat details. These are classified under the threat category `[Suspicious]` and threat family `[Suspicious\_C2C]`. This represents an active command-and-control event where a single external source IP (`103.10.24.44`) targeted a specific internal victim IP (`192.168.28.59`).

6. Log Storage and Processing Inefficiency: Exactly 50% of the dataset (42,225 rows) consists of duplicate rows. This duplication doubles the storage requirements and processing bandwidth, leading to inflated operational costs for log management and SIEM ingestion.

Opportunities

7. Firewall Rule Optimization: Transitioning traffic from the broad `Allow-any` rule to specific, application-aware rules (such as the existing `Allow-Web-Browsing` rule) presents an opportunity to significantly harden the network's security posture.

8. Log Cost Reduction: Resolving the log duplication issue will immediately reduce log storage and SIEM licensing costs by up to 50%.

Recommendations

Based on the statistical and business analysis, the following actions are recommended:

1. Immediate Threat Mitigation:

1. Isolate and conduct an immediate forensic investigation on the internal host at IP `192.168.28.59`.
2. Block all inbound and outbound traffic to the external malicious IP `103.10.24.44` associated with the 34 flagged `[Suspicious\_C2C]` command-and-control events.

2. Firewall Policy Hardening:

3. Audit and restrict the broad `Allow-any` rule (which allowed 63,321 connections).
4. Migrate allowed traffic to more specific, application-aware rules to enforce the principle of least privilege.



3. Logging Infrastructure Correction:

5. Investigate the syslog forwarding and firewall logging configurations to identify and resolve the root cause of the 42,225 duplicate rows.

4. Address Data Quality Limitations:

6. Establish longer-term logging windows to overcome the 20-minute temporal limitation of this dataset, allowing for baseline trend analysis.

7. Investigate why 99.96% of threat-related fields are empty to ensure threat intelligence feeds are properly integrated and logging details are fully captured.

Conclusion

The analysis of the Palo Alto PA-440 firewall logs for Demo-IR-Org reveals a highly active network dominated by standard networking and DNS traffic. However, the presence of 34 targeted command-and-control (C2C) security events on a specific internal host highlights an active threat that must be addressed immediately. Furthermore, significant operational inefficiencies—specifically an over-permissive `Allow-any` rule and a 50% log duplication rate—expose the organization to unnecessary security risks and inflated log management costs. Implementing the recommended rule audits, host isolation, and logging corrections will significantly improve both the security posture and operational efficiency of the organization.